

Ê PHASOR TEMPLATES (Air: ε_r=μ_r=1)

Wavenumber per medium <i>nonmag.</i>			
$k_i = \frac{\omega\sqrt{\epsilon_{r,i}}}{c}$ (rad/m) $\Rightarrow k_2 = k_1\sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$			
Direction table $u \in \{x, y, z\}$ = axis perp. to boundary			
Inc dir	Inc	Refl	Trans
+û	e^{-jk_1u}	e^{+jk_1u}	e^{-jk_2u}
-û	e^{+jk_1u}	e^{-jk_1u}	e^{+jk_2u}

+û direction ⇔ Med 2 at u ≥ 0. Reflected sign flips, transmitted matches incident.

General template (incident has pol. ê, amplitude a ₀):	
$\tilde{\mathbf{E}}^i = a_0 \hat{e} e^{\mp j k_1 u}$ (V/m)	(incident wave, Med 1)
$\tilde{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm j k_1 u}$ (V/m)	(reflected wave, Med 1)
$\tilde{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp j k_2 u}$ (V/m)	(transmitted wave, Med 2)
(lossy med 2: replace $e^{\mp j k_2 u}$ with $e^{\mp \alpha_2 u} e^{\mp j \beta_2 u}$)	
Total in medium 1: $\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r$.	
ê by polarization plug into templates above	

Pol.	ê	Note
Lin. x̂	ŷ	E _y = 0
Lin. ŷ	ŷ	E _x = 0
Lin. at 45°	(x̂ + ŷ)/√2	in phase
RHC	x̂ - jŷ	δ = -π/2
LHC	x̂ + jŷ	δ = +π/2
General	a _x x̂ + a _y e ^{jδ} ŷ	elliptical

Γ, τ act on ê unchanged. Only swap the polarization, not the formulas.

TABLE 8-2 — Γ, τ, R, T (unitless) SUMMARY

<i>Med. 1 (z < 0) incident side; Med. 2 (z ≥ 0) transmitted.</i>			
	⊥ pol (θ _i = 0)	∥ pol	
Γ refl. wav	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ trans. wav	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ rel.	τ = 1 + Γ	τ _⊥ = 1 + Γ _⊥	τ _∥ = (1 + Γ _∥) $\frac{\cos \theta_i}{\cos \theta_t}$
R refl. pwr	Γ ²	Γ _⊥ ²	Γ _∥ ²
T trans. pwr	τ ² $\frac{\eta_1}{\eta_2}$	τ _⊥ ² $\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$	τ _∥ ² $\frac{2 \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
R + T total	T = 1 - R	T _⊥ = 1 - R _⊥	T _∥ = 1 - R _∥

Notes: sin θ_t = √μ₁ε₁/μ₂ε₂ sin θ_i; nonmag. ⇒ η₂/η₁ = n₁/n₂.

Intrinsic impedance η _i plug into table above	
$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)}$	η ₀ = 120π ≈ 377 Ω
Default μ _r = 1 (air, dielectric). Use μ _r ≠ 1 only if problem says “ferromagnetic”/iron/ferrite/etc.	
Low-loss correction η _c σ/(ωε) ≪ 1	
$\eta_c \approx \sqrt{\frac{\mu}{\epsilon}} \left(1 + j \frac{\sigma}{2\omega\epsilon}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$	
For Γ/τ just use η ₀ /√ε _r . The j term is for η _c , θ _η . See LOSSY MEDIA — 5 CASES.	

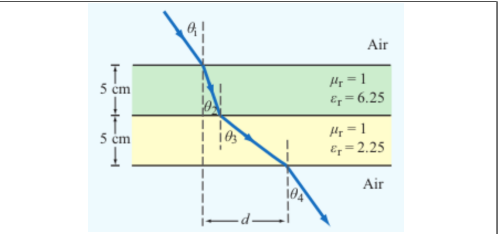
POWER REFLECTION & TRANSMISSION

% reflected <i>always</i> %P _r = 100 Γ ²
% transmitted <i>η-ratio matters!</i> %P _t = 100 τ ² η ₁ /η ₂
Check: %P _r + %P _t = 100.

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary <i>angles from normal</i>	$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$
Multilayer chain <i>stack of slabs</i> 1 → 2 → 3 → 4	$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$
Air-slabs-air shortcut <i>outer media identical</i>	θ _{exit} = θ _{incident}
<i>Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).</i>	

LATERAL DISPLACEMENT



Beam offset through N slabs thickness t_i, refracted angle inside slab i Slab-indexed θ_i = angle inside slab i; t_i = slab thickness (m)

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

The angle (θ_i) in the formula is always the one the ray makes inside that slab of height (t_i).

LOSSY MEDIA — 5 CASES

Loss tangent ε' = ε _r ε ₀ ; ε ₀ = 8.854 × 10 ⁻¹² F/m		
	ε''/ε'	σ/ωε ₀
Type	σ/ωε	η _c , α, β
Lossless (σ = 0)	0	η = η ₀ /√ε _r exact α = 0, β = ω√ε _r /c
Low-loss	< 10 ⁻²	η _c ≈ $\frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma \eta_0}{2\omega\epsilon_r}\right)$ α ≈ $\frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}$, β ≈ $\frac{\omega\sqrt{\epsilon_r}}{c}$
Quasi-cond.	10 ⁻² –10 ²	exact: (η/√X)e ^{jθ_η} see below
Good cond.	> 10 ²	η _c ≈ (1 + j)√πfμσ α ≈ β ≈ √πfμσ
Magnetic	any	keep full √μ _r /ε _r η ₀

For Γ/τ: drop j correction; use η₀/√ε_r. Magnetic: keep μ_r.

Low-loss quick params σ/(ωε) ≪ 1, μ _r = 1	
α ≈ $\frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}$ (Np/m), β ≈ $\frac{\omega\sqrt{\epsilon_r}}{c}$ (rad/m), η _c ≈ $\frac{\eta_0}{\sqrt{\epsilon_r}}$ (Ω)	
Exact (quasi-cond) X = √[1 + (ε''/ε') ²]	
α = ω√ $\frac{\mu\epsilon'}{2}$ √X-1, β = ω√ $\frac{\mu\epsilon'}{2}$ √X+1	
θ _η = $\frac{1}{2} \arctan \frac{\sigma}{\omega\epsilon'}$	

RECTANGULAR WAVEGUIDE — MODES

TE mode <i>E_z = 0, H_z ≠ 0</i>	TM mode <i>H_z = 0, E_z ≠ 0</i>
<i>Dimensions a × b with a > b along x̂, ŷ; wave in ẑ.</i>	
<i>E_x form (TE) read m, n from arguments</i>	
$E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$	
<i>coef. on x = mπ/a; coef. on y = nπ/b.</i>	

Identify TE_{mn} vs TM_{mn} cos/sin pattern of E_x is identical for both!

1. Read the problem statement (e.g. “a TE wave” → TE).
2. If m = 0 or n = 0 in the mode label ⇒ must be TE (TM forbids 0).
3. Source field H_z(x, y) given ⇒ TE. Source field E_z(x, y) given ⇒ TM.

TE allows at least one of m, n ≥ 1 (other can be 0). TM needs both m, n ≥ 1. So TE₁₀ exists but TM₁₀ doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

e^{-jβz} factor omitted. k_c² = (mπ/a)² + (nπ/b)². Ê in V/m, Ĥ in A/m, Z in Ω.

TE mode <i>generator: Ê_z</i>	$\tilde{H}_z = H_0 \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$
$\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{E}_z = 0, \tilde{H}_x = -\tilde{E}_y/Z_{TE}, \tilde{H}_y = \tilde{E}_x/Z_{TE}$	
$Z_{TE} = \eta / \sqrt{1 - (f_c/f)^2}$	

TM mode <i>generator: Ê_z</i>	$\tilde{E}_z = E_0 \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$
$\tilde{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{H}_z = 0, \tilde{H}_x = -\tilde{E}_y/Z_{TM}, \tilde{H}_y = \tilde{E}_x/Z_{TM}$	
$Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$	

TEM plane wave *for reference*
 $\tilde{E}_x = E_0 e^{-j\beta z}, \tilde{H}_y = \tilde{E}_x/\eta$

$$\eta = \sqrt{\mu/\epsilon} \text{ (}\Omega\text{)}, f_c \text{ N/A, } k = \omega\sqrt{\mu\epsilon}$$

Properties common to TE/TM	
$f_c = \frac{u_{p0}}{2} \sqrt{(m/a)^2 + (n/b)^2} \text{ (Hz)}$	
$\beta = k\sqrt{1 - (f_c/f)^2} \text{ (rad/m)}$	
$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_c/f)^2}} \text{ (m/s)}$	

CUTOFF FREQUENCY

Common cutoffs (a > b) f _c formula in Table 8-3	
Mode	f _{mn}
u _{p0}	c/√ε _r (hollow: u _{p0} = c)
TE ₁₀	f ₁₀ = u _{p0} /(2a) (dominant)
TE ₀₁	f ₀₁ = u _{p0} /(2b)
TE ₂₀	f ₂₀ = u _{p0} /a = 2f ₁₀
TE ₁₁ , TM ₁₁	f ₁₁ = $\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$

Propagation requires f > f_{mn}. In Z_{TE}/Z_{TM} and u_g, f_c = f_{mn} of the excited mode.

ε_r FROM DISPERSION

Given ω, β, mode (m, n) <i>result is unitless</i>	
$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$ (unitless)	
<i>Inputs: ω (rad/s), β (rad/m), a, b (m), c = 3 × 10⁸ (m/s), m, n (unitless).</i>	

GROUP VELOCITY & TRAVEL TIME

$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} \text{ (m/s), } u_p u_g = u_{p0}^2$	
t = L/u _g (s) travel time through guide of length L	
<i>Higher modes have lower u_g ⇒ arrive later.</i>	

BOUNDARY

η _i , η _t — imp. (Ω)
Γ, τ, R, T — coeffs (unitless)
τ = 1 + Γ; R = Γ ²
k _i = ω√ε _{r,i} /c (rad/m)
α ₂ (Np/m), β ₂ (rad/m): lossy
z = 0 — boundary plane (m)
η ₀ = 120π ≈ 377 Ω

SNELL / OBLIQUE

θ _i , θ _t — angles from normal
n _i = √ε _{r,i} — index of refr.
t _i — slab thickness (m)
d — lateral disp. (m)
plane of inc. = plane of k+normal
⊥: E ⊥ this plane
∥: E ∥ this plane

WAVEGUIDE

a, b — wide, narrow dim. (m);
a > b
m, n — mode indices (int.)
f _{mn} — cutoff (Hz)
β — prop. const (rad/m)
Z _{TE} /TM — wave imp. (Ω)
E _x — E-field comp. (V/m)
H _y — H-field comp. (A/m)
η = η ₀ /√ε _r

VELOCITIES

u _{p0} = c/√ε _r — bulk phase vel.
u _p = u _{p0} /√[1 - (f _c /f) ²]
u _g = u _{p0} √[1 - (f _c /f) ²]
u _p u _g = u _{p0} ²
t = L/u _g — travel time
c = 3 × 10 ⁸ m/s

POWER & UNITS

%P _r , %P _t — pwr refl./trans. (unitless)
%P _r = 100 Γ ²
%P _t = 100 τ ² η ₁ /η ₂
%P _r + %P _t = 100
σ (S/m), ε (F/m), μ (H/m)
ε ₀ = 8.85 × 10 ⁻¹² F/m
μ ₀ = 4π × 10 ⁻⁷ H/m

DIPOLE TYPE BY LENGTH

STEP 0 — always start here λ = wavelength (m), l = antenna rod length (m), $c=3\times 10^8$ m/s

$\lambda = \frac{c}{f}$ (m)	ratio $\frac{l}{\lambda} \rightarrow$	pick row below
l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D=1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave shortcut, $D=1.64$, $R_{\text{rad}}=73\ \Omega$
$= 1$	Full-wave	ARB, $D=2.41$, $R_{\text{rad}}=199\ \Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — wrong shape.

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors small $l \ll \lambda$
$\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta$, $\vec{H}_\phi = \vec{E}_\theta / \eta_0$
Power density $l/\lambda < 1/50$; far field
$S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2 \theta = S_{\text{max}} \sin^2 \theta$ (W/m ²)
l rod length (m); I_0 peak current (A); R obs. dist. (m); θ from dipole axis; $k=2\pi/\lambda$ (rad/m); $\eta_0=120\pi \approx 377\ \Omega$.
Max @ broadside $\theta_{\text{max}} = \pi/2$; D exact $D=1.5$ (1.76 dB)
Total radiated power $P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l/\lambda)^2$ (W)

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at R any l
$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$
At $\theta = 0, \pi$: bracket is 0/0. L'Hôpital $\Rightarrow S=0$ (no axial radiation). TI-36X Pro alt: evaluate bracket at $\theta = 0.001$ rad $\rightarrow$ tiny $\rightarrow 0$.
Normalized pattern <i>dimensionless</i> $F(\theta) = S(\theta)/S_{\text{max}}$
Quarter-wave $l = \lambda/4$ $\pi l/\lambda = \pi/4$, $\cos(\pi/4) = \sqrt{2}/2$
$\frac{S(\theta)}{S_0} = \left[\frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$, $S_0 = \frac{15I_0^2}{\pi R^2}$
$S_{\text{max}} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858 S_0$
$D=1.64$, $R_{\text{rad}}=36.5\ \Omega$, $\theta_{\text{max}} = \pi/2$.
Half-wave $l = \lambda/2$ $\pi l/\lambda = \pi/2$, <i>shortcut form</i>
$S(\theta) = \frac{15I_0^2}{\pi R^2} \frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta}$
$D=1.64$, $R_{\text{rad}}=73\ \Omega$, $\theta_{\text{max}} = \pi/2$.

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

Conversion <i>multiply by $\pi/180$ or $180/\pi$</i>
$\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$ $\theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$
° rad sin θ cos θ sin ² θ cos ² θ
0 0 0 1 0 1
30 $\pi/6$ 1/2 $\sqrt{3}/2$ 1/4 3/4
45 $\pi/4$ $\sqrt{2}/2$ $\sqrt{2}/2$ 1/2 1/2
60 $\pi/3$ $\sqrt{3}/2$ 1/2 3/4 1/4
90 $\pi/2$ 1 0 1 0
180 π 0 -1 0 1

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY

Setup N identical elements along axis, spacing d
elem pos $z_i = id$, $i = 0, \dots, N-1$; $A_i = a_i e^{j\psi_i}$
Equal phase ($\psi_i=0$) $\Rightarrow$ broadside beam ($\theta_{\text{max}} = \pi/2$).
Progressive phase $\psi_i = i\alpha \Rightarrow$ steered beam at $\cos \theta_{\text{max}} = -\alpha/(kd)$.
Pattern multiplication S_e = single element pattern
$S(R, \theta, \phi) = S_e(R, \theta, \phi) F_a(\theta)$
Array factor (general) $k=2\pi/\lambda$
$F_a(\theta) = \left \sum_{i=0}^{N-1} A_i e^{jkid \cos \theta} \right ^2$
Uniform $A_i=1$, equal phase <i>closed form</i>
$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$, $\gamma = kd \cos \theta$
Peak: $F_a^{\text{max}} = N^2$ at $\gamma=0$ (broadside, $\theta = \pi/2$). Normal-ize: $F_a^{\text{norm}} = F_a/N^2$.
HPBW (broadside, uniform) <i>use Nd, not $(N-1)d$</i>
$\beta \approx 0.88 \frac{\lambda}{\sqrt{Nd}}$ (rad) = $50.8^\circ \cdot \frac{\lambda}{\sqrt{Nd}}$
Grating lobes appear if $d > \lambda$ (broadside). Avoid.

DIPOLE

l — antenna length (m)
$\lambda = c/f$ (m); $c=3\times 10^8$ m/s
I_0 — peak current (A)
$k=2\pi/\lambda$ (rad/m)
R — obs. dist. (m); θ from axis
$\eta_0=120\pi \approx 377\ \Omega$
$S(R, \theta)$ — pwr density (W/m ²)

BEAM

$F(\theta) = S/S_{\text{max}}$ (unitless)
a — coef. on θ^2 in Gaussian
I_0 — HPBW (rad or °)
Ω_p — pattern solid angle (sr)
$D=4\pi/\Omega_p$ — directivity
ξ — rad. efficiency; $G=\xi D$
$D_{\text{dB}} = 10 \log_{10} D$; $D = 10^{D_{\text{dB}}/10}$

FRIS / dB

P_t — power transmitted (W)
P_r — power received (W)
G_t — transmit gain (linear)
G_r — receive gain (linear)
$S_r = G_t P_t / (4\pi R^2)$ (W/m ²)
$P_r = G_t G_r (\lambda / 4\pi R)^2 P_t$
$G=10^{G_{\text{dB}}/10}$
3 dB = $\times 2$, 10 dB = $\times 10$
$P_N = k_B T_{\text{sys}} B$ (W)

APERTURE

A_p — physical area (m ²)
$A_e = \lambda^2 D / (4\pi)$ (m ²)
$A_e \approx A_p$ for aperture
$D \approx 4\pi A_p / \lambda^2$ (unitless)
$\beta \approx 0.88 \lambda / \ell$ (rect/sq.)
$\beta \approx 1.02 \lambda / d$ (circular)
$2A_p \Rightarrow 2D$, $\beta/\sqrt{2}$

ARRAY

N — number of elements
d — spacing (m)
$\gamma = kd \cos \theta$
$F_a = \sin^2(N\gamma/2) / \sin^2(\gamma/2)$
$F_a^{\text{max}} = N^2$ at $\theta = \pi/2$
$F_a^{\text{norm}} = F_a/N^2$
$\beta \approx 0.88 \lambda / (Nd)$ (broadside)

BEAM PARAMETERS — β , Ω_p , D

Normalize <i>always start here</i>
$F(\theta) = S(\theta)/S_{\text{max}}$ (unitless)
Beamwidth β at threshold X
Set $F(\theta_X) = X$, solve for θ_X ; $\beta = 2\theta_X$ (symmetric)
Threshold $X=10^{\text{dB}/10}$ θ_X (Gauss. short-cut)
HPBW (−3 dB) 0.5 $\sqrt{\frac{\ln 2}{a}}$
−6 dB 0.25 $\sqrt{\frac{\ln 4}{a}}$
−10 dB 0.1 $\sqrt{\frac{\ln 10}{a}}$
any $F(\theta_X)$ any X $F(\theta_X) = X$
Pattern solid angle Ω_p <i>recognize pattern, look up</i>
$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$ (sr)
No ϕ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta \, d\theta$.
θ is the integration variable — DON'T plug in a number for θ (e.g. $\theta_{1/2}$). Integrate over $\theta \in [0, \pi]$.
Pattern $F(\theta)$ Ω_p (sr)
Isotropic ($F=1$) $4\pi \approx 12.57$
Hertzian sin ² θ $8\pi/3 \approx 8.38$
Half-wave dipole ≈ 7.66
Narrow Gauss $e^{-a\theta^2}$ $\pi/a = \pi\theta_{1/2}^2 / \ln 2$
2-axis HPBW (aperture) $\beta_{xz} \cdot \beta_{yz}$
Other (use integral) numerical

TI-36X Pro: [2nd] [d/dx] for $\int_0^\pi$, RAD mode, multiply by 2π .

Directivity D <i>always $D=4\pi/\Omega_p$; common shortcuts below</i>
$D = \frac{4\pi}{\Omega_p}$ (unitless) $D_{\text{dB}} = 10 \log_{10} D$
WORD TRAP: “direction of max radiation” = angle θ_{max} (rad/deg). “directivity” = unitless number D . Different things!
Dipoles $\rightarrow$ COMMON DIPOLE PARAMS table. Aperture: $D \approx 4\pi A_p / \lambda^2$. 2-axis HPBW: $D \approx 4\pi / (\beta_{xz} \beta_{yz})$. Circular: $D \approx 4\pi / \beta^2$.
Gain (if asked) $\xi = \text{rad. eff.}$, $0 < \xi \leq 1$
$G = \xi D$ $G_{\text{dB}} = 10 \log_{10} G$
Lossless / ideal antenna $\Rightarrow G=D$.

COMMON DIPOLE PARAMETERS

For Friis, $G_t = G_r = D$ below (lossless antennas).
Type l D D_{dB} R_{rad} (Ω)
Isotropic N/A 1 0 —
Hertzian $< \lambda/50$ 1.5 1.76 $80\pi^2 (l/\lambda)^2$
Half-wave $\lambda/2$ 1.64 2.15 73.2
Monopole $\lambda/4$ (gnd) 1.64 2.15 36.6
Full-wave λ 2.41 3.82 199

Half-wave: $P_{\text{rad}} = 36.6 I_0^2$ W. Monopole ($\lambda/4$ above ground): same patt. as half-wave but P_{rad} , R_{rad} halved. Lossless antenna ($\xi=1$) $\Rightarrow G=D$. So in Friis, use $G_t, G_r =$ these D values (e.g. half-wave $G=1.64$).

ANTENNA AREAS (A_e , A_p)

Effective area <i>antenna's RX cross section; D from BEAM PARAMS or table above</i>
$A_e = \frac{\lambda^2 D}{4\pi}$ (m ²)
Physical cross section <i>wire dipole, diameter d; l see table above</i>
$A_p = l d$ (m ² , any dipole)
Inverse: D from A_e
$D = \frac{4\pi A_e}{\lambda^2}$ (unitless)
Hertzian special case $D=1.5$
$A_e^{\text{Hertz}} = \frac{3\lambda^2}{8\pi} \approx 0.119 \lambda^2$ (m ²)

Thin wire: $A_e \gg A_p$. Antenna captures field over $\sim \lambda$.

FRIS TRANSMISSION FORMULA

Pwr density at RX, then RX pwr G_t, G_r linear gains; $\lambda = c/f$

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{); } P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \text{ (W)}$$

Equivalent: $P_r = S_r A_{e,r}$.

Solve for R *max range*

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using A_e *recv. area form*

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.

General form (off-axis) *when patterns NOT peak-aligned*

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

F_t, F_r normalized patterns at the link directions; $F=1$ when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert any dB gain to linear: $G=10^{G_{\text{dB}}/10}$.

3. Get G_t, G_r by antenna type:

- named dipole $\rightarrow$ COMMON DIPOLE PARAMS ($G=D$ lossless)
- aperture / dish $\rightarrow$ APERTURE ANTENNAS ($G=D \approx 4\pi A_p / \lambda^2$)
- arbitrary pattern $F(\theta) \rightarrow$ BEAM PARAMS ($G=\xi \cdot 4\pi/\Omega_p$)
- given in dB $\rightarrow$ dB $\leftrightarrow$ LINEAR table

4. Plug into Friis.

Use linear G , not dB. Convert FIRST.

dB $\leftrightarrow$ LINEAR

Works for any power-like quantity ($G, D, P_r/P_t, \text{SNR}, \dots$):

$X_{\text{dB}} = 10 \log_{10} X_{\text{lin}}$	$X_{\text{lin}} = 10^{X_{\text{dB}}/10}$
dB linear	dB linear
0 1	10 10
3 2	13 20
6 4	20 100
7 5	30 1000
−3 0.5	−10 0.1

Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use $20 \log$ not $10 \log$ (power $\propto E^2$).

NOISE & SNR

Noise power <i>thermal noise into matched receiver</i>
$P_N = k_B T_{\text{sys}} B$ (W)
$k_B = 1.38 \times 10^{-23}$ J/K; T_{sys} system noise temp (K); B receiver bandwidth (Hz).
Signal-to-noise ratio
$\text{SNR} = \frac{P_{\text{rec}}}{P_N}$ (unitless) $\text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$
Typical comm. link needs SNR > 10 dB for usable signal.

APERTURE ANTENNAS (horn, dish)

A_p physical aperture area (m ²); A_e effective area (m ² , $= \lambda^2 D / 4\pi$); β_{xz}, β_{yz} HPBW's in two planes (rad); ℓ, d aperture side / diameter (m).
Directivity (any shape, uniform illum.)
$D \approx \frac{4\pi A_p}{\lambda^2}$ (unitless) $\Rightarrow A_e \approx A_p$
Directivity from beamwidths <i>any aperture</i>
$D \approx \frac{4\pi}{\beta_{xz} \beta_{yz}}$ (use rad); circular: $D \approx 4\pi / \beta^2$

HPBW and D by aperture shape <i>uniform illum.; β in rad</i>
Shape A_p HPBW (rad) D (unitless)
Rect. $\ell_x \times \ell_y$ $\beta_x \approx 0.88 \lambda / \ell_x$ $D \approx 4\pi / (\beta_x \beta_y)$
ℓ_y $\beta_y \approx 0.88 \lambda / \ell_y$ $= 4\pi \ell_x \ell_y / \lambda^2$
Square ℓ ℓ^2 $\beta \approx 0.88 \lambda / \ell$ $D \approx \frac{4\pi}{\beta^2} = \frac{4\pi \ell^2}{\lambda^2}$
Circular (diam. d) $\pi d^2/4$ $\beta \approx 1.02 \lambda / d$ $D \approx \frac{4\pi}{\beta^2} = \frac{\pi^2 d^2}{\lambda^2}$
Scaling rules $D \propto A_p / \lambda^2$; $\beta \propto \lambda / \sqrt{A_p}$
Change New D New β
Double area $D_{\text{new}} = 2D_{\text{old}}$ $\beta_{\text{new}} = \beta_{\text{old}} / \sqrt{2}$ ($A_p \rightarrow 2A_p$)
Double freq ($\lambda \rightarrow D_{\text{new}} = 4D_{\text{old}}$ $\beta_{\text{new}} = \beta_{\text{old}}/2$ ($\lambda/2$)